Guidelines for the Description of Maize Germplasm Resources

1 Scope

This specification establishes the descriptors for corn germplasm resources and their grading criteria.

These guidelines apply to the collection, organization, and preservation of maize germplasm resources; the development of data standards and data quality control specifications; and the establishment of database and information-sharing network systems.

2 Normative References

The provisions of the following documents shall form part of this specification by reference therein. For dated reference documents, all subsequent corrigenda (excluding errata) or revised editions shall not apply to this specification; however, parties to agreements concluded in accordance with this specification are encouraged to consider whether the latest editions of those documents may be used. For undated reference documents, the latest editions shall apply to this specification.

ISO 3166 Codes for the Representation of Names of Countries

GB/T 2659: Country and Region Name Codes

GB/T 2260 Administrative Divisions Code of the People's Republic of China

GB/T 12404 Code for Unit Affiliation

GB/T 3543.1-7—1995: Code for Inspection of Crop Seeds

GB 1353—1999 Corn

GB/T 8613—1999 Corn for the Starch Fermentation Industry

NY/T 520—2002 High-quality protein maize

NY/T 521—2002 High-oil corn

NY/T 597—2002 High-starch corn

NY/T 523—2002 Sweet Corn

NY/T 524—2002 Starchy Corn

NY/T 522—2002 Popcorn

3 Terms and Definitions

3.1 corn

Gramineae; Maydeae; genus Zea L.; maize varieties; annual herbaceous plants; scientific name *Zea mays*

L.; also known as bao gu, bao mi, bang zi, yu jiao, zhen zhu mi, etc.; chromosome number 2n=2x=20. Maize plants are relatively tall and consist of roots, stems, leaves, male ears, female ears, and other parts (Figure 1).

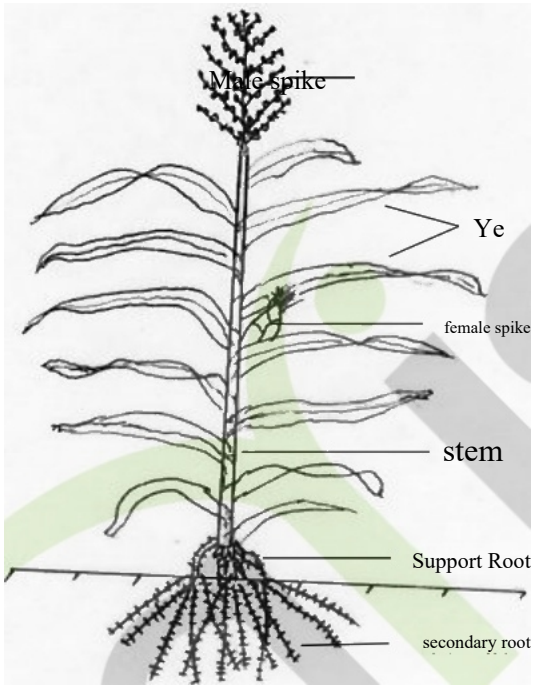


Figure 1: Maize Plant

3.2 Corn Germplasm Resources

Wild maize resources, local varieties, composite varieties/synthetic populations, selfing lines, genetic materials, etc.

3.3 Essential Information

Basic descriptive information on maize germplasm resources, including the national unified identifier, germplasm name, scientific name, place of origin, germplasm type, and other details.

3.4 Morphological Characteristics and Biological Properties

Phenological stages, botanical morphology, yield traits, and other characteristic traits of maize germplasm resources.

3.5 Quality Characteristic

Nutritional quality traits of maize germplasm resources, including protein content, fat content, starch content, lysine content, and other parameters.

3.6 Stress Resistance

The adaptability or resistance of maize germplasm resources to various abiotic stresses, including drought tolerance, cold tolerance, waterlogging tolerance, high-density tolerance, and salt tolerance.

3.7 Resistance to pests and diseases

The adaptability or resistance of maize germplasm resources to various biotic stresses, including large spot disease, Fusarium leaf spot, stem rot, ear rot, damping-off, dwarf mosaic disease, and corn borer, among others.

3.8 combining ability

The concept of combining ability emerged during research into the genetic basis of heterosis and the development of breeding methods for maize hybrids. Combining ability refers to the productivity or the magnitude of other trait indicators produced by the first-generation hybrids resulting from the cross between one parental line and another parental line; it is not manifested through the traits of the parents themselves, but is instead estimated from the mean phenotypic value of the target trait (e.g., yield) expressed in the hybrids formed by that parental combination. There is a distinction between general combining ability and specific combining ability. When a particular selfing line or variety is crossed with several other selfing lines or varieties, the mean value of the yield or a specific other trait exhibited by the first-generation hybrids is termed its general combining ability (GCA). GCA reflects the general relationship between that parental line and many other parental lines; its magnitude is determined by the number of favorable gene loci present in the parental line, is controlled by additive genetic effects, and is a heritable component. Parents with high general combining ability have a greater likelihood of producing high-yield hybrids; consequently, general combining ability has become an important indicator in the selection of parental lines for maize breeding. When a particular selfing line or variety is crossed with another particular selfing line or variety, the yield or other trait value exhibited by the F1 hybrid is referred to as the specific combining ability (SCA) of that particular combination. SCA reflects the specific relationship between two particular parents, is determined by non-additive genetic effects, and is controlled by gene dominance, superdominance, and epistatic effects; it can only be manifested in specific hybrid combinations through interactions between alleles or non-allelic interactions of the parents and is therefore nonheritable. The identification of combining ability described in this specification refers to the assessment of heritable general combining ability.

4 Essential Information

4.1 Nationwide Unified Number

The unique identifier for germplasm. The unified national numbering system for corn germplasm resources consists of a 2-digit germplasm type code, followed by a 2-digit provincial or municipal code and a 4-digit sequence number for domestic resources; for foreign resources, it consists of a 2-digit germplasm type code, a 1-digit continental code, a 2-digit country code, and a 3-digit sequence number. These identifiers are uniformly assigned by the lead institution responsible for the cataloging of corn germplasm resources.4. 2. Germplasm Bank Numbering

The accession number assigned to maize germplasm resources in the National Long-term Crop Germplasm Resource Bank consists of "I1G" followed by a five-digit sequential number.

4.3 Introduction Number

The designation assigned to maize germplasm when it was introduced from abroad.

4.4 Collection ID

The identification number assigned to maize germplasm during its field collection.

4.5 Germplasm Name

The Chinese name of domestic germplasm or the Chinese translation of imported germplasm from abroad.

4.6 Foreign Name of Germplasm

The foreign-language name of the introduced germplasm or the Chinese Pinyin name of the domestic germplasm.

4.7 Field Name

Grasses (Gramineae).

4.8 Generic Name

The maize genus (*Zea* L.).

4.9 scientific name

The scientific name of corn is *Zea mays L.*

4.10 country of origin

The name of the country of origin, the name of the region, or the name of the international organization of the maize germplasm.

4.11 Country of origin

Name of the province of origin for domestic maize germplasm; name of the national first-level administrative region of origin for imported foreign germplasm.

4.12 Country of Origin

The names of the original counties, townships, and villages of domestic maize germplasm.

4.13 height

Altitude of the original habitat of the maize germplasm. The unit is m.

4.14 longitude

The longitude of the origin of maize germplasm is expressed in degrees (°) and minutes (′). The format is DDDFF, where DDD represents degrees and FF represents minutes.

4.15 latitude

The latitude of the original origin of maize germplasm, expressed in degrees (°) and minutes (′). The format is DDFF, where DD represents degrees and FF represents minutes.

4.16 Origin

For foreignly introduced maize germplasm: the name of the country of origin, the name of the region, or the name of the international organization; for domestic germplasm: the name of the province or county of origin.

4.17 Preserving entity

The name of the original institution where the maize germplasm was preserved prior to its submission to the National Long-term Crop Germplasm Resource Bank.

4.18 Save Unit Number

The germplasm accession number assigned by the original germplasm preservation institution.

4.19 Lineage

The genetic relationships among self-crossing lines derived from maize breeding, as well as those among integrated varieties/synthetic populations.

4.20 Breeding Unit

The name of the institution or individual responsible for the development of self-crossed maize lines and integrated varieties/synthetic populations.

4.21 Age at maturation

Years in which self-crossed lines and integrated varieties/synthetic populations were successfully developed through maize breeding.

4.22 Breeding Method

Breeding methods for maize inbred lines and composite varieties/synthetic populations.

4.2 3 types of quality

Corn germplasm types are divided into six categories.

Industrial wild resources

2 local variety

3 Comprehensive varieties/synthetic populations

4 Inbred line

5 genetic stocks

6 other

4.24 picture

Image filename for maize germplasm. The image format is.jpg.

4.25 observation place

Name of the observation site for the morphological characteristics and biological traits of maize germplasm.

5 Morphological Characteristics and Biological Properties

5.1 Sowing Time

The sowing date for the identification of the morphological characteristics and biological traits of maize germplasm. It shall be expressed as "year month day," in the format "YYYYMMDD."

5.2 period of emergence

The date on which more than 50% of the young shoots in the entire area emerge from the soil surface at a depth of 3.0 cm or greater. This shall be expressed as "year month day" in the format "YYYYMMDD."

5.3 Mating period

The date on which the tip of the male spike of more than 50% of the plants in the entire district emerges above the terminal leaves. It shall be expressed as "year month day," in the format "YYYYMMDD."

5.4 Flowering and Pollen Dispersal Stage

The date on which the male spikes of more than 50% of the plants in the entire district begin to flower and release pollen. It shall be expressed as "year month day," in the format "YYYYMMDD."

5.5 Silk-Spinning Period

The date on which the filaments of the female panicles of more than 50% of the plants in the entire district emerge from the bracts. It shall be expressed as "year month day," in the format "YYYYMMDD."

5.6 Maturation Period

In the entire district, over 90% of the grains in the plants enter the full maturation stage from the late wax stage; the milk line in the grains at the middle of the panicle disappears; the grains dehydrate and harden; a black layer appears at the base; and the husks turn yellow. This date shall be expressed as "year month day" in the format "YYYYMMDD."

5.7 Days from sowing to emergence

The number of days elapsed from the day following sowing until the emergence stage. The unit is d.

5.8 Days from seedling emergence to male flower emergence

The number of days elapsed from the seedling emergence stage to the male flower emergence stage. The unit is d.

5.9 Number of days from pollination to flowering and pollen release

The number of days elapsed from the male flower emergence stage to the flowering and pollen dispersal stage. Units: d.

5.10 Number of days from flowering and pollen release to silk emergence

The number of days elapsed from the flowering and pollen dispersal stage through the silk emergence stage; the unit is d.

5.11 Number of Days from Silk Production Initiation to Maturity

The number of days elapsed from the silk-spinning stage to maturity. The unit is d.

5.12 number of growing days

The number of days elapsed from the day following sowing until maturity. The unit is d.

5.13 Effective accumulated temperature (≥10°C)

Cumulative temperature above 10 °C from the day after sowing until maturity; expressed in °C. 5.14 Planting density

Number of seedlings established per hectare. Unit: plants/hm².

5.15 Seedling leaf color

The color of the first leaf of maize during the seedling stage.

1 Light green

2 green

3 Deep green

4 purple

5.16 Seedling bud sheath color

Observe the color of the bud sheath after the first leaf has unfurled.

1 green

2 Green band with purple streaks

3 purple

4 Deep purple

5.17 Seedling vigor

The vigor of the plants during the seedling stage.

1 weak

2 China

3 Strong

5.18 Branching capacity

The tillering capacity of the germplasm.

0 not have

1 weak

2 China

3 Strong

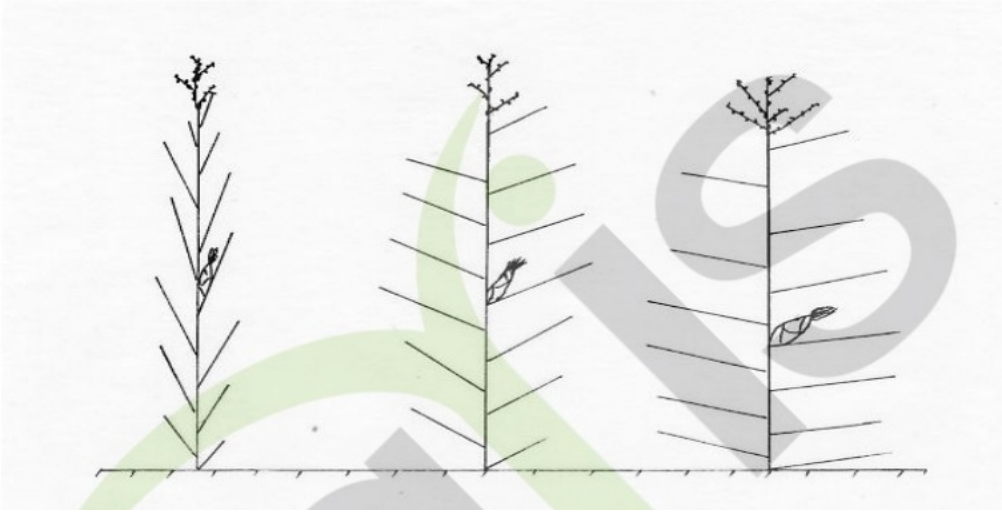
5.19 plant type

This is determined by the angle between the leaf and the main stem, also known as the leaf attachment angle (see Figure 2).

1 compact

2 intermedius type

3 Downy type



1 2 3

Figure 2: Plant Architecture

5.20 Number of first-order branches of the male spike

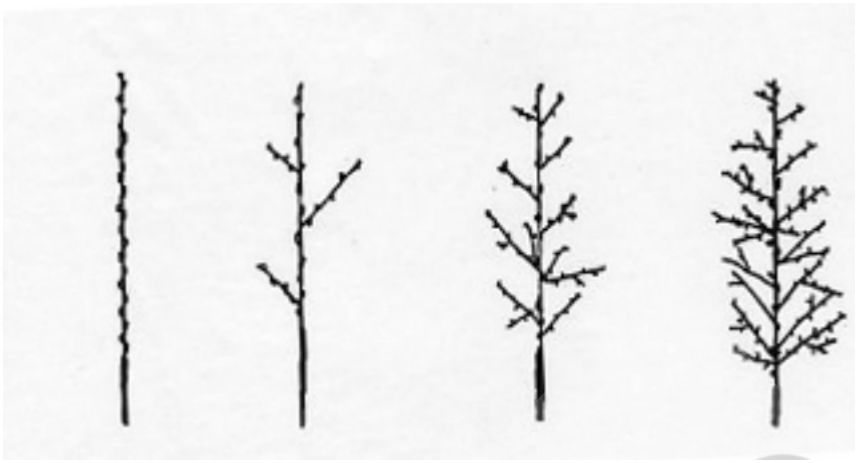
The number of branches on the main axis of the male inflorescence (see Figure 3).

0 not have

1 few

2 China

3 More



0 1 2 3

Figure 3: Primary Branches of the Male Spike

5.21 Long Male Spike

Length of the main axis of the male inflorescence. Unit: cm.

5.22 Color of the Male Spike Sheath

The color of the male spike sheath.

Green Industry

2 Green band with purple streaks

3 purple

5.23 Anther color

Color of the anthers in the male spikelet.

1 green

2 lilac colour

3 purple

4 Deep purple

5.24 Filament color

Color of the female spike florets.

1 Yellow-green

2 light red

3 Deep red

4 Variegated

5.25 plant height

The height of the main stem from the ground to the top of the male spike. Unit: cm.

5.26 ear height

The height from the ground to the node at which the main stem attaches to the uppermost effective female spike (a spike that has normally matured and produced more than 10 fruits; the same applies below). The unit is cm.

5.27 Number of leaves on the main stem

The total number of leaves borne on the main stem of the plant. The unit is leaves.

5.28 Number of leaves at the upper ear

The number of leaves on the effective female spike at the uppermost part of the main stem of the plant.

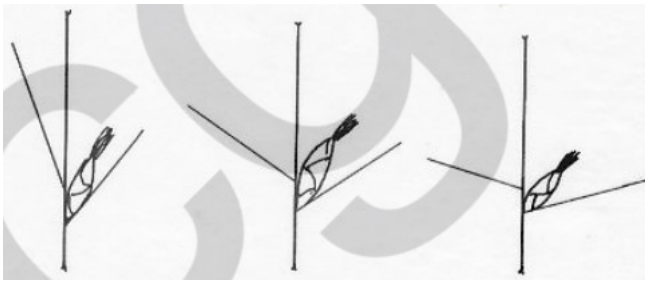
5. Angle of the upper leaves on the 29th upper panicle

The growth angle of the first leaf above the position of the effective female spike at the uppermost part of the main stem of the plant (see Figure 4).

1 up-right

2 Middle

3 Flat



worker 2 3

Figure 4: Angle of the Upper Leaves on the Main Spike

5.30 Leaf Color of the Upper Panicle Leaves

The color of the first leaf above the position of the effective female spike at the uppermost part of the main stem of the plant.

1 ondine

2 green

3 Deep green

5.31 Leaf length of the upper panicle leaves

The length from the leaf tongue to the leaf tip of the first leaf above the position of the effective female spike at the uppermost part of the main stem of the plant. Unit: cm.

5.32 Leaf width of the upper panicle leaves

The width at the midpoint of the first leaf above the position of the effective female spike at the uppermost part of the main stem of the plant. Unit: cm.

5.33 Support Root Development Degree

Development of the supporting roots in the upper part of the main stem above ground.

0 not have

1 weak

2 China

3 Strong

5.34 Plant uniformity

The uniformity of growth and development among plants.

1 in good order

2 Relatively uniform

3 irregularity

5.35 Number of effective tillers per plant

The number of stems on a single plant of the tillering germplasm that develop into effective female spikes, excluding the main stem; expressed in units.

5.36 Single stem effective spike count

The number of effective female spikes developed from a single stem. Unit: number.

5.37 Double Spike Rate

The percentage of plants bearing two or more valid female spikes on a single stem relative to the total number of plants, expressed as a percentage (%).

5.38 Empty stalk rate

The percentage of plants in the population with zero effective female spikes relative to the total number of plants, expressed as a percentage (%).

5.39 stem diameter

The average diameter at the midpoint of the third internode of the aboveground stem of a typical plant (excluding the leaf sheath), expressed in cm.

5.40 Downing rate

The percentage of plants in the mature population that have fallen over at the root (tilt angle> 45°) relative to the total number of plants, expressed as a percentage (%).

5.41 Backlash Rate

The percentage of plants in the mature stage whose stems have bent backward relative to the total number of plants. Expressed as a percentage (%).

5.42 Integrity of the female spike sheath

The condition of the female panicle enclosed by bracts (see Figure 5).

1 Fully encapsulated

2 Incomplete encapsulation



2

Figure 5: Condition of the Female Spike Sheath

5.43 Stalk Length

The length from the base of the female spike to the point of attachment to the main stem. Unit: cm.

5.44 Spikelet stalk angle

The state in which the panicle stalk of the female panicle extends outward is expressed by the angle it makes with the main stem. The unit is (°).

5.45 Fresh Weight per Spike

Fresh weight of the total number of effective female spikes per plant, expressed in g.

5.46 Moisture content of mature grains

The moisture content of fresh grains measured at the time of maturity harvest, expressed as a percentage (%).

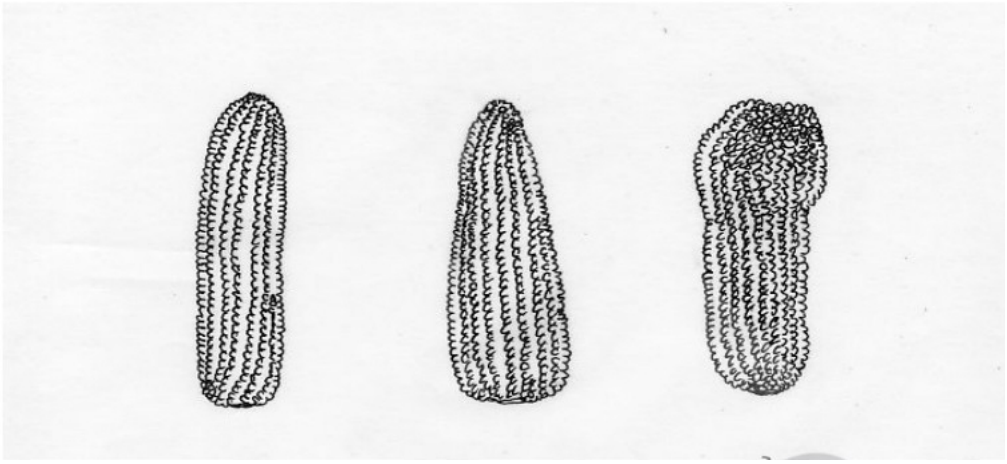
5.47 Spike Shape

The shape of the female spike of the germplasm (see Figure 6).

1 cylindricality

2 taper

3 Flattened



1 2 3

Figure 6: Spike-Shaped

5.48 穗 Length

Length from the base to the tip of the female spike. Unit: cm.

5.49 Long, Pointed Tip

Length of the non-grain portion at the tip of the female spike. Unit: cm.

5. 50 穗粗

Diameter of the middle portion of the female spike; for flattened spikes, measure the short diameter. Unit: cm.

5.51 Row Count

The number of grains per female spike ranges over an even number.

5.52 Number of particles

Number of grains per row in the female spike. Unit: grains.

5.53 granules

Grain trait types after maturation (see Figure 7).

Hard granular type

2 Toothed type

3 intermedius type

4 Sticky type

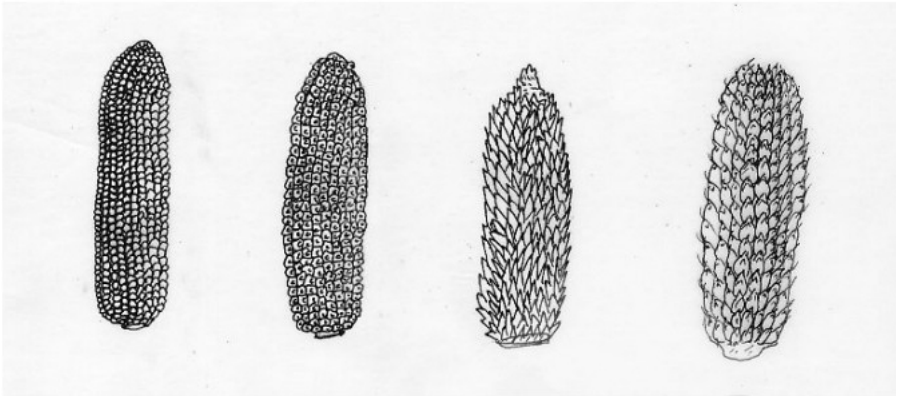
5 Sweet-type

6 Explosive type

7 Powdery type

8 Has a raft-type configuration

9 Sweet-pale type



worker 2 6 8

Figure 7: Representative Particle Morphology

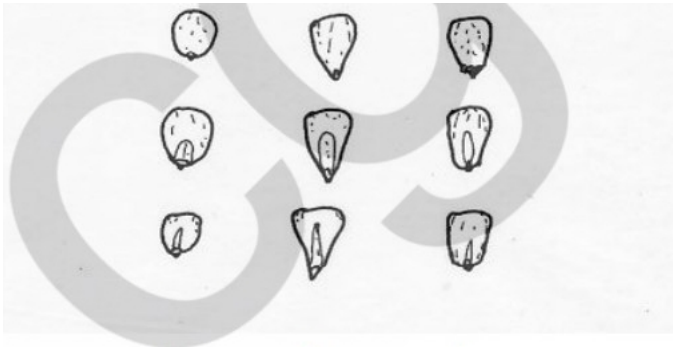
5.54 Granule shape

The shape of the grains after maturation (see Figure 8).

1 circular

2 wedge

3 Intermediate form



1 2 3

Figure 8. Grain Shape

5.55 Granule size

Grain size after maturation.

1 Small-grain type

2 Medium-grain type

3 Large-grain type

5.56-colored granules

http://www.cgris.net

The appearance and color of the grains after maturation.

1 white
2 pale yellow
3 Huang
4 orange
5 orange
6 red
7 Lan
8 purple
9 black
10 leukotrichia
11 yellow blood streaks
12 Yellow grains with white filaments
13 Yellowish erythema
14 Top white
15 design and colour
16 White/Yellow
17 Yellow/White
18 Yellow/Red
19 Red/Yellow
20 Yellow/Purple
21 Purple/Yellow
22 Variegated

5.57 axial colour

Color of the rachis of the mature female spike.

1 white
2 pale red
3 red
4 purple

5 White/Red

6 Red/White

5.58 Thick shaft

After maturation, the diameter at the midpoint of the female spike axis; for flattened spike axes, measure the short diameter. Units: cm.

5.59 Single plant spike dry weight

Dry weight of the total number of effective female spikes per plant (naturally air-dried; grain moisture content is assumed to be 13%; the same applies below). The unit is g.

5.60 Single-Plant Grain Dry Weight

Dry weight per single grain. Unit: g.

5.61 Seed yield rate

The percentage of grain dry weight relative to panicle dry weight, expressed as a percentage (%).

5.62 thousand seeds per weight

The weight of 1,000 intact grains measured by random sampling (grain moisture content assumed to be 13%); expressed in g.

5.63 Granule bulk density

Weight of grains per liter of volume (grain moisture content assumed to be 13%); the unit is g/L.

5.64 Grain yield

Hectare yield equivalent to the dry weight of individual grains from sampled plots; the unit is kg/hm².

5.65 Fresh Weight of Biological Yield

The hectare yield calculated from the total fresh weight of the aboveground parts of the entire plant in the plot measured at the time of corn harvest. The unit is kg/hm².

5.66 Dry biomass yield

The hectare yield equivalent to the total dry weight of the entire plant in the above-ground portion of the sampling plot. The unit is kg/hm².

5.67 General coordination ability

The average value of yield or another specific trait exhibited by the F1 hybrid resulting from the cross between a particular genotype and multiple self-crossed lines or varieties. It is divided into three types.

1 weak

2 China

3 Strong

6 quality characteristic

6.1 Total starch content

The percentage of starch contained in the grain relative to the grain's mass, expressed as a percentage (%).

6.2 Amylose content

The percentage of amylose in the granules relative to the total starch mass, expressed as%.

6.3 Amylopectin content

The percentage of amylopectin in the endosperm relative to the total starch mass, expressed as%.

6.4 Soluble sugar content

The percentage of soluble sugars in the grain relative to the total sugar content of the grain, expressed as a percentage (%).

6.5 Crude protein content

The percentage of protein content relative to the mass of the grain, expressed as a percentage (%).

6.6 Lysine content

The percentage of lysine in the kernel relative to the total protein content of the kernel, expressed as a percentage (%).

6.7 Crude fat content

The percentage of fat content relative to the mass of the grain, expressed as%.

6.8 Fatty Acid Value (KOH)

The number of milligrams of potassium hydroxide required to neutralize the free fatty acids in 100 g of kernels and their products. The unit is 10^2 mg/g.

6.9 Vitamin E Content

The percentage of vitamin E contained in the grain relative to the grain's mass, expressed as a percentage (%).

6.10 Explosion rate

The percentage of kernels that burst upon heating and form corn kernels relative to the total number of kernels, expressed as a percentage (%).

6.11 Extrusion ratio

The ratio of the volume of corn kernels after they have expanded and burst into popcorn to the volume of the kernels, expressed in multiples.

6.12 Eating Quality

One of the quality indicators for fresh, sweet, and sticky corn is a quality assessment made after manual tasting, based on a comprehensive evaluation of the sample's sweetness, viscosity, husk thickness, flavor, mouthfeel, and other attributes. It is divided into three categories.

1 Excellent

2 China

3 difference

7 Stress resistance

7.1 Drought Tolerance During the Budding Stage

The germination capacity of maize seeds under drought stress.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.2 Drought tolerance during the seedling stage

The ability of maize seedlings to tolerate or resist drought stress during the seedling stage.

Work: Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.3 Drought tolerance during the flowering and pollen dispersal stage

The ability of maize to tolerate or resist drought stress during the flowering and pollen dispersal stage.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.4 Drought tolerance during the grouting period

The ability of maize to tolerate or resist drought stress during the grain-filling stage.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.5 Cold hardiness during the budding stage

The germination capacity of maize seeds under low-temperature stress.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.6 Cold tolerance during the seedling stage

The ability of corn seedlings to tolerate or resist low temperatures during the seedling stage.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.7 Cold resistance during the grouting period

The ability of corn to tolerate or withstand low temperatures during the grain-filling stage.

1 Extremely strong

3 **Strong**

5 China

7 weak

9 Extremely weak

7.8 salt tolerance

The ability of maize germplasm to tolerate or resist salt stress.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.9 Waterlogging tolerance during the seedling stage

The tolerance or resistance of maize germplasm seedlings to waterlogging during the seedling stage.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

7.10 Density resistance

The ability of maize germplasm to tolerate high planting density.

1 Extremely strong

3 Strong

5 China

7 weak

9 Extremely weak

8 Disease and Pest Resistance

8.1 Large spot disease resistance

The resistance of maize to the disease caused by the large-spot turgid umbilicate fungus [*Exserohilum turcicum* (Pass.) Leonard et Suggs], whose sexual stage is the large-spot setosphaeric fungus *Setosphaeria turcica* (Luttrell) Leonard et Suggs, primarily characterized by the formation of large lesions on the leaves.

Work High Resistance (HR)

3 disease-resistant(R)

5 Zhong Kang (MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.2 Resistance to spot disease

The resistance of maize to a disease caused by the maize pathogen *Bipolaris maydis (* Nisikado et Miyake) Shoemaker, whose sexual stage is the heterostrophic cavity fungus *Cochliobolus heterostrophus (* Drechsler) Drechsler, primarily characterized by small lesions on the leaves.

highly resistance(HR)

3 Disease-Resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.3 Resistance to *Coryneum* leaf spot

The resistance of maize to a disease primarily caused by fungi of the genus Curvularia, predominantly *Curvularia lunata* (Walk.) Boed., whose sexual stage is *Cochliobolus lunatus* Nalson et Haasis, characterized mainly by the formation of small lesions on the leaves.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.4 Resistance to gray mold

The resistance of maize to diseases caused by fungi of the genus *Cercospora*, such as * Cercospora zeamaydis* (Tehon & Daniels), which primarily manifest as leaf lesion symptoms.

1 highly resistance(HR)

3 disease-resistant(R)

5 Zhong Kang (MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.5 Round spot disease resistance

The resistance of maize to the disease caused by *Bipolaris zeicola* [*Bipolaris zeicola* (Stout) Shoemaker], whose sexual stage is *Cochliobolus carbonum* *Cochliobolus carbonum* Nelson, primarily manifests as leaf lesion symptoms.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.6 Resistance to Scab

The resistance of maize to the disease caused by *Physoderma maydis* [*Physoderma maydis* (Miyabe) Miyabe], which primarily manifests as brown spots on the stems and leaves.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.7 Rust resistance

The resistance of maize to common rust and southern rust, caused respectively by Puccinia sorghi *Puccinia sorghi* Schwein. and Puccinia polysora Underw., which primarily manifest as rust spots on the leaves and stems.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.8 μm resistance to smut

The resistance of maize to the disease caused by Sphacelotheca *reiliana* (Kühn) Clinton, which primarily manifests as blackhead symptoms on the female and male ears.

High resistance (HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.9 Resistance to powdery mildew

The resistance of maize to the disease caused by the maize smut fungus [Ustilago *zeae (* Beckm.) Unger], which primarily manifests as black lesions on the female ears, stems, and leaves.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.10 Resistance to wood rot

Resistance to diseases primarily characterized by chlorosis of stem leaf sheaths caused by fungi of the genus Rhizoctonia, including Rhizoctonia solani Kühn, whose sexual stage is Thanatephorus cucumeris (Frank) Donk], and Rhizoctonia zeae [*Rhizoctonia zeae* Voorhees, whose sexual stage is *Weitea circinata* Warcup et Talbot], among others.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.11 Stem rot resistance

The resistance of maize to diseases primarily caused by fungi of the genus Pythium spp. and the genus Fusarium *Fusarium spp.*, which result in stem-base rot leading to wilt or yellow wilt symptoms throughout the plant.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.12 Resistance to stem blight

The resistance of maize to diseases primarily caused by Fusarium fungi, such as *Fusarium verticilloides* (Sacc.) Nirenberg = ***F. moniliforme*** Sheldon, whose sexual stage is *Gibberella fujikuroi* (Sawada) Wollenw., as well as fungi of the genus Aspergillus spp., characterized mainly by symptoms of rotting in the grains of the female panicles.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.13 Resistance To疯顶病

The resistance of maize to the disease caused by *Sclerophotora *macrospora* (Saccado) Thirum. Shaw et Naras.], which primarily manifests as abnormal growth of the male and female ears.

Work High Resistance (HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.14 Resistance to coarse shiv

The resistance of maize to the disease caused by maize rough dwarf virus (*Maize rough dwarf virus)*, which primarily manifests as whole-plant dwarfism and thickened internodes.

1 highly resistance(HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.15 Resistance to dwarf mosaic disease

The resistance of maize to the disease caused by the Sugarcane mosaic virus–maize dwarfism B strain (*Sugarcane mosaic virus–Maize dwarf* strain B), which primarily manifests as whole-plant dwarfism and leaf mosaic symptoms.

1 High resistance (HR)

3 disease-resistant(R)

5 moderate resistance(MR)

7 Sick state (S)

9 High Sensitivity (HS)

8.16 Corn Borer Resistance

The resistance of maize to the primary symptoms of the Asian corn borer [*Ostrinia* *furnacalis (* Guenée)] characterized by leaf feeding infestation.

1 highly resistance(HR)

3 Insect-resistant (R)

5 moderate resistance(MR)

7 Insect Sensitivity (S)

9 High Sensitivity (HS)

9 Other characteristics

9.1 fertility

In the process of sexual reproduction, maize has the ability to produce normal male and female gametes and to produce normal offspring. The inability to produce normal female ears, filaments, or female gametes, or the inability to produce normal anthers or pollen, or the inability of male and female gametes to combine properly to produce offspring, is referred to as infertility; this can be classified as female sterility or male sterility. In maize, hybrid varieties can be developed using male sterile lines.

1 Fertile

2

9.2 Male Sterility Types

Types of male sterility control in male sterile germplasm.

1 Nuclear infertility

2 Cytoplasmic sterility

3 Nuclear–cytoplasmic interaction-induced sterility

9.3 Sterile Cell Cytoplasmic Clade

A group of male sterile germplasm under cytoplasmic control.

1 Tgroup

2 Cgroup

3 Sgroup

9.4 Karyotype

A formula representing the number, size, morphology, and structural characteristics of chromosomes.

9.5 DNAfinger-print

DNA fingerprint profiles of each corn germplasm resource identified using molecular marker-based methods.

9.6 use

Primary utilization pathways for maize germplasm resources.

1 edible

2 For feed

3 industrial

9.7 remarks

A detailed description of any special descriptors or special codes for the corn germplasm that have not been covered above.